Live cell-based assay for detection of CNS autoantibodies

This protocol cannot be used for diagnostic purposes.

Reagents

- *Cells*
  - GripTite™ 293 MSR Cell Line, R79507, Invitrogen, derived 293-H cells, expressessing the human macrophage scavenger receptor type 1 (MSR I).
- *Plastic hardware*
  - ibiTreat microscope-grade flat bottom 96-well plates, 89626, Ibidi/Micromedic
  - 75 ml tissue-culture flasks
  - Sterile 1.5 ml Eppendorf tubes
  - Suitable pipette tips, etc
- *Chemical and biological substances*
  - Poly-D-Lysine, 0.1 mg/mL, A3890401, ThermoFisher
  - Dulbecco's Modified Eagle Medium, high glucose (DMEM), D6429-500ML, Merck
  - OptiMEM reduced serum medium, 31985070, ThermoFisher
  - Dulbecco's Phosphate-Buffered Saline, D8537-24X500ML, Merck
  - Fetal bovine serum (FBS) - heat inactivation not necessary
  - Bovine serum albumin (BSA)
  - MEM Non-Essential Amino Acids (NEAA), 11140050, ThermoFisher
  - Geneticin, 10131035, ThermoFisher
  - HEPES, H3375-500 g, Merck
  - Penicillin, Streptomycin/Amphotericin-B, 100x concentrated, 15240062, ThermoFisher
  - Complete medium: DMEM with 10% FBS, 1% NEAA, 600 µg/ml geneticin
  - NMDA-R staining medium: Complete medium without FBS but with 1% BSA, 0.5% HEPES and 1% Penicillin, Streptomycin/Amphotericin-B
  - Versene, 15040066, ThermoFisher
  - Trypsin-1 mM EDTA, 25200056, ThermoFisher
  - Trypan blue, 15250061, ThermoFisher
  - Lipofectamine 2000 kit, 11668019, ThermoFisher
  - Lipofectamine 3000 kit, L3000015, ThermoFisher
  - Goat anti-Human IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 568, A21090, ThermoFisher
  - Goat anti-human IgG Fc cross-adsorbed secondary antibody, 31125, Invitrogen
  - Donkey anti-goat IgG (H+L) Alexa fluor 568, A11057, Invitrogen
  - Formaldehyde solution, 16%, 28908, ThermoFisher
  - Sodium bisulphite solution, 13438-2.5L-R, Merck
  - Fluoromount-G with DAPI, 00-4959-52, ThermoFisher
- *Plasmids*
  - CASPR2-EGFP plasmid, VB230730-1331huu, VectorBuilder
    - Bulk-acquired from VectorBuilder
  - GABAB-R1-3xFLAG plasmid, VB220705-1040shn, VectorBuilder
    - Plasmid-containing bacteria bought from VectorBuilder, expanded and purified locally for higher purity.
  - GABAB-R2-3xFLAG plasmid, VB220705-1041ads, VectorBuilder
    - Plasmid-containing bacteria bought from VectorBuilder, expanded and purified locally for higher purity.
  - NMDA-R plasmid (GRIN1, untagged) and LGI1-EGFP: kind gifts from Dr Patrick Waters, Autoimmune Nerology Diagnostic Laboratory, University of Oxford, UK.
    - Added to competent bacteria, expanded and purified locally.
  - IgLON5-GFP plasmid, VB230614-1124geq, VectorBuilder
    - Bulk-acquired from VectorBuilder
  - MOG-3xFLAG plasmid, VB230614-1134mfv, VectorBuilder.
    - Plasmid-containing bacteria bought from VectorBuilder, expanded and purified locally for higher purity.

Continuously: Maintenance and splitting of 293 MSR cells

1. Grow MSR Cell Line in a T75 flask containing 10 mL of complete medium
2. Cells are split twice a week:
    3. Remove media from the flask
    4. Rinse the flask with 6 ml of D-PBS
    5. Add 1 ml of pre-warmed Versene to the flask
    6. Incubate for 5 minutes at room temperature
    7. Decant Versene and add 1 ml of pre-warmed, 0.25% trypsin-1 mM EDTA solution to the flask
    8. Incubate at 37°C until cells have detached (about 1 minute)
    9. Add 4 ml of prewarmed complete medium
    10. Transfer the cell suspension to a 15 ml sterile, conical tube
    11. Centrifuge the cells at 100 g for 10 minutes
    12. Resuspend in 2 mL of prewarmed complete medium
    13. Add 200 µL of the cell suspension to 10 mL of prewarmed complete medium
    14. Transfer to a new T75 flask
    15. Change the cells at least every 30 passages to limit phenotypic drifting

Day 1: coating of plates with Poly-D-Lysine (PDL)

1. Dilute Poly-D-Lysine to 25 µg/ml in sterile PBS.
2. Add 75 µL of the diluted Poly-D-Lysine per well
3. Incubate for 24 h at room temperature under sterile conditions.
4. Flick off remaining solution
5. Rinse with 200 µL PBS three times
6. Dry for 2 hours
7. After this, the plates are ready to use

Day 2: seeding the 293 MSR cells

1. Follow steps 3-12 for the cell line splitting protocol
2. Transfer 10 µL of the cell suspension to an Eppendorf tube
3. Add 10 µL Trypan blue solution
4. Count the cells
5. Add 30.000 cells in 150 µL of complete medium to each used well in the 96-well plate

16. Incubate in a humified incubator at 37° C, 5% CO_2 overnight

Day 3: Transfection

Table 1, amounts per well

Antigen	Amount of plasmid(s), ng	Lipofectamine kit	Plus/P3000 reagent, uL
CASPR2	150	2000	0.75
GABAB-R	376 (GABABR1), 150 (GABABR2)	3000	1.5
IgLON5	200	3000	1
LGI1	300	3000	1.5
MOG	150	2000	0.75
NMDA-R (GRIN1)	420	2000	1.5

1. For each well
    a. Mix plasmid (amounts in Table 1), **AND** Plus reagent **OR** Reagent P3000 (reagent and amount depending on antigen, see Table 1)
    b. Add OptiMEM to top up to a total volume of 7.5 µL per used well
2. For each well, mix 1.5 µL of lipofectamine 2000 **OR** 3000 (depending on antigen) with 6 µL of OptiMEM
3. Mix the plasmid and lipofectamine mixtures and incubate at room temperature for 10 minutes
4. Add 15 µL of the mixture to each used well without changing the medium that the cells were originally in

5. **For NMDA-R only**: after 8 hours, the transfection media was replaced with 150 µL of fresh complete medium with the NMDA-R inhibitor MK801 at a 30 µM concentration.
6. **For NMDA-R only**: stain after another 16 hours, i.e. 24 hour transfection time before staining
7. **For all antigens except for NMDA-R**: 48 hour transfection time before staining

Day 4-5 - Labeling with antibodies

1. Invert plate on a clean tissue, and transfer to a second clean tissue to dry it completely
2. Add patient sera as well as negative and positive controls diluted 1:20 (1:100 for CASPR2) in 100 µL DMEM + 1% NEAA
    a. For NMDA-R stainings, 1% HEPES and 1% BSA should also be added to the medium.
    b. One sample or control to each well.
3. Incubate 45 min at room temperature
    a. For NMDA-R stainings, 60 minutes is recommended.
4. Invert plate on a clean tissue, and transfer to a second clean tissue to dry it completely
5. **For NMDA-R only**: Add 150 µL DMEM+0.5% HEPES to each well and invert plate on a clean tissue. Repeat once.
6. Add 200 µL PBS to each well
7. Invert plate on a clean tissue
8. Add 100 µL formaldehyde, 4% diluted in PBS
9. Incubate for 10 min at room temperature in externally ventilated hood
10. Flip the plate upside-down to empty the formaldehyde in a container filled with tissues, to minimise splashing
11. Add 200 µL PBS to each well
12. Flip the plate upside-down to empty the formaldehyde-contaminated PBS in the same tissue-containing container
    a. Inactivate the PFA in the soaked tissues with enough 4% W/V sodium bisulfite solution in PBS to have at least a 1:1 ratio to the formaldehyde volume.
    b. This leads to hydroxymethylsulphonate (HMS) formation, which is biologically inactive and can be discarded as normal waste.

Protocol fork:

For all antigens except for NMDA-R

13. Add 100 µL AF568-conjugated goat anti-human IgG in PBS. Incubate for 45 minutes at room temperature.
    1. Dilute 1:400 for GABAB, IgLON5 and MOG, 1:600 for CASPR2
14. Incubate for 45 minutes at room temperature.
15. Invert plate on a clean tissue, and transfer to a second clean tissue to dry it completely
16. Add 200 µL PBS
17. Invert plate on a clean tissue, and transfer to a second clean tissue to dry it completely
18. Add two drops of DAPI-containing mounting medium
19. Acquire images on microscope

For NMDA-R only

13. Add 150 µL DMEM+0.5% HEPES and invert plate on a clean tissue
14. Add 100 µL unconjugated goat anti-human antibody diluted 1:500 in DMEM+0.5% HEPES+1%BSA
15. Incubate for 45 minutes at room temperature
16. Add 150 µL DMEM+0.5% HEPES and invert plate on a clean tissue
17. Add 100 µL AF568-conjugated Donkey anti-goat IgG diluted 1:500 in DMEM+0.5% HEPES+1%BSA.
18. Incubate for 45 minutes at room temperature.
19. Add 150 µL DMEM+0.5% HEPES and invert plate on a clean tissue. Repeat once.
20. Add 200 µL PBS
21. Invert plate on a clean tissue, and transfer to a second clean tissue to dry it completely
22. Add two drops of DAPI-containing mounting medium
23. Acquire images on microscope